

Chapter 1, Problem 4RQ

Problem

A charge of 2 C flowing past a given point each second is a current of 2 A.

- (a) True
- (b) False

Step-by-step solution

Step 1 of 2

Write mathematical relationship between current i , charge q , and time t is,

$$i \triangleq \frac{dq}{dt}$$

Step 2 of 2

Amount of charge $dq = 2 \text{ C}$.

The time for which the charge is flowing, $dt = 1 \text{ s}$.

Substitute 2 C for dq and 1 s for dt in the mathematical equation.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{2}{1} \\ &= 2 \text{ A} \end{aligned}$$

Thus, statement that "a charge of 2C flowing past a given point each second is a current of 2A" is **true**.